

Comparative Analysis of Unsupervised and Supervised Autoencoders for Nuclei Classification in Clear Cell Renal Cell Carcinoma Images

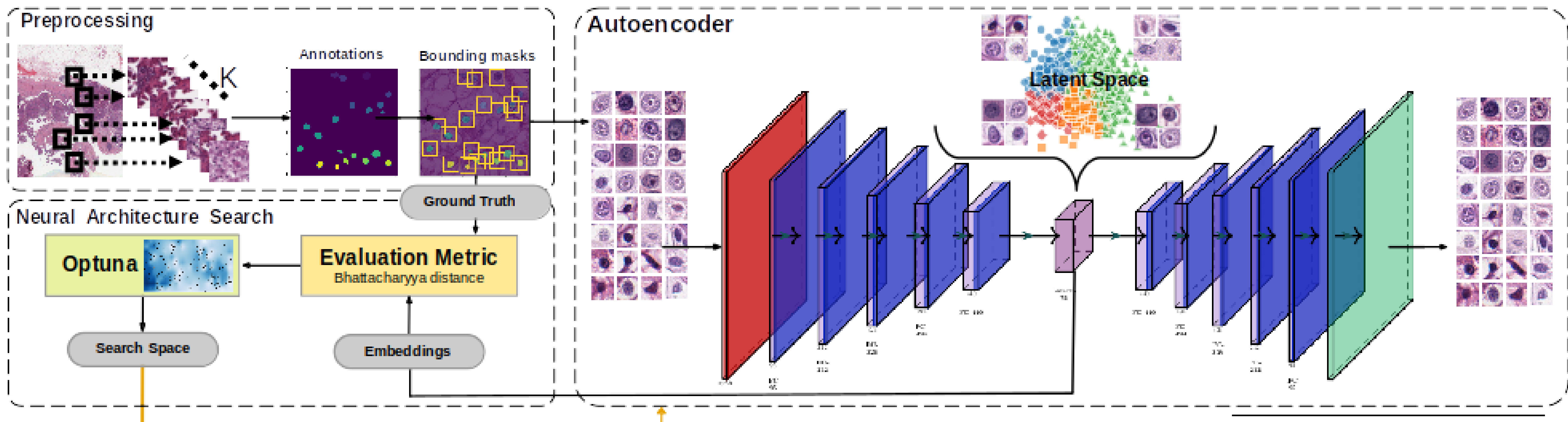
Fatemeh Javadian¹ Zahra Aminparast² Johannes Stegmaier¹ Abin Jose¹

¹Institute of Imaging and Computer Vision, RWTH Aachen University, Aachen, Germany

²Institute of Medical Sciences, Kermanshah University, Medical School, Kermanshah, Iran

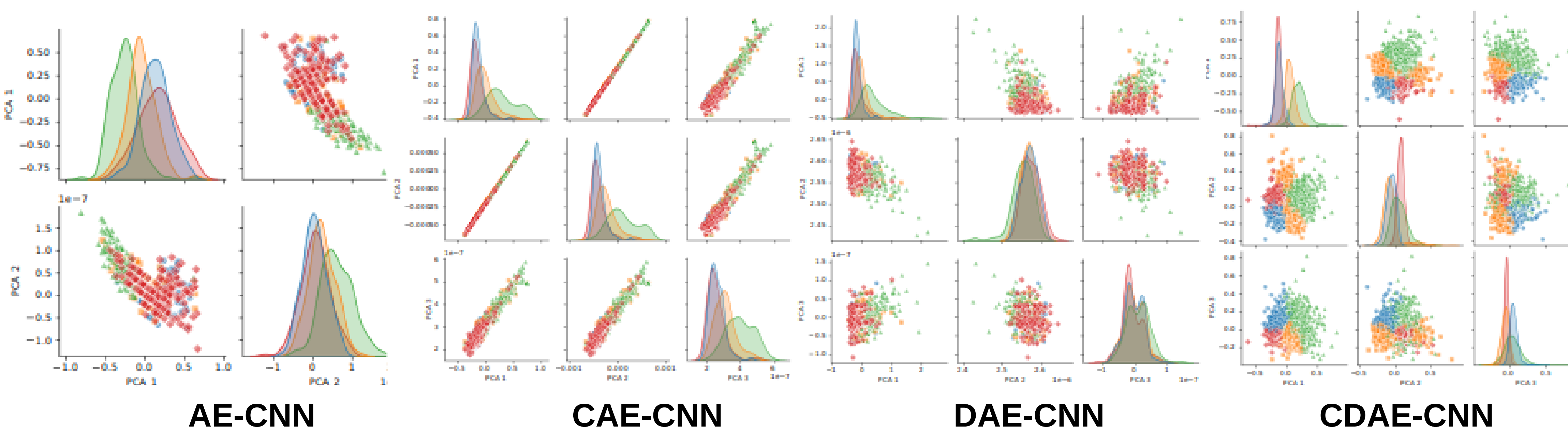


Poster No. 1571091608 - 5



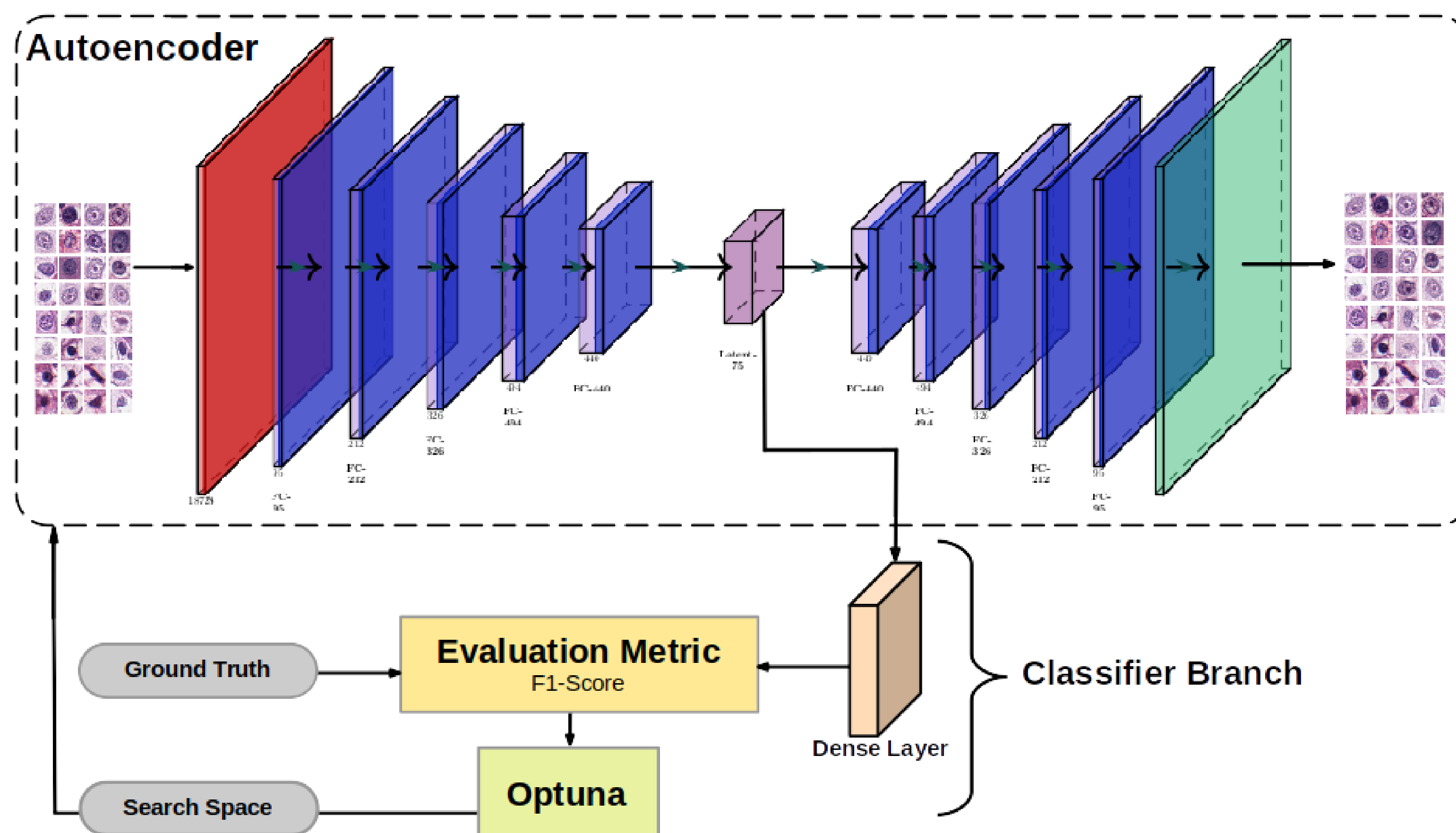
INTRODUCTION

Grading nuclei in clear cell renal cell carcinoma (ccRCC) is essential for accurate cancer diagnosis, traditionally requiring detailed manual inspection by pathologists. This study investigates supervised and unsupervised **autoencoder architectures** to automate **nuclei classification**, aiming to enhance grading accuracy and reduce reliance on manual annotations.



EXPERIMENTS & RESULTS

We evaluated four autoencoder variants (**AE**, **CAE**, **DAE**, **CDAE**) optimized via **Optuna** and assessed using **Bhattacharyya distance** and **F1-score** metrics. Latent space analysis demonstrated superior class separability for the **supervised model (CDAE)**, particularly when optimized by Bhattacharyya distance. Further optimization targeting classification performance (F1-score) improved classification metrics, despite reduced latent space separability.



Metric	Bhattacharyya Opt.	F1 Opt.
Overall Precision	0.6104	0.6985
Overall Recall	0.6111	0.7008
Overall F1	0.6113	0.6994
Grade 1 F1	0.5571	0.6373
Grade 2 F1	0.4469	0.5481
Grade 3 F1	0.6890	0.7821
Non-tumorous F1	0.7520	0.8300
Bhattacharyya dis.	47.23	36.90

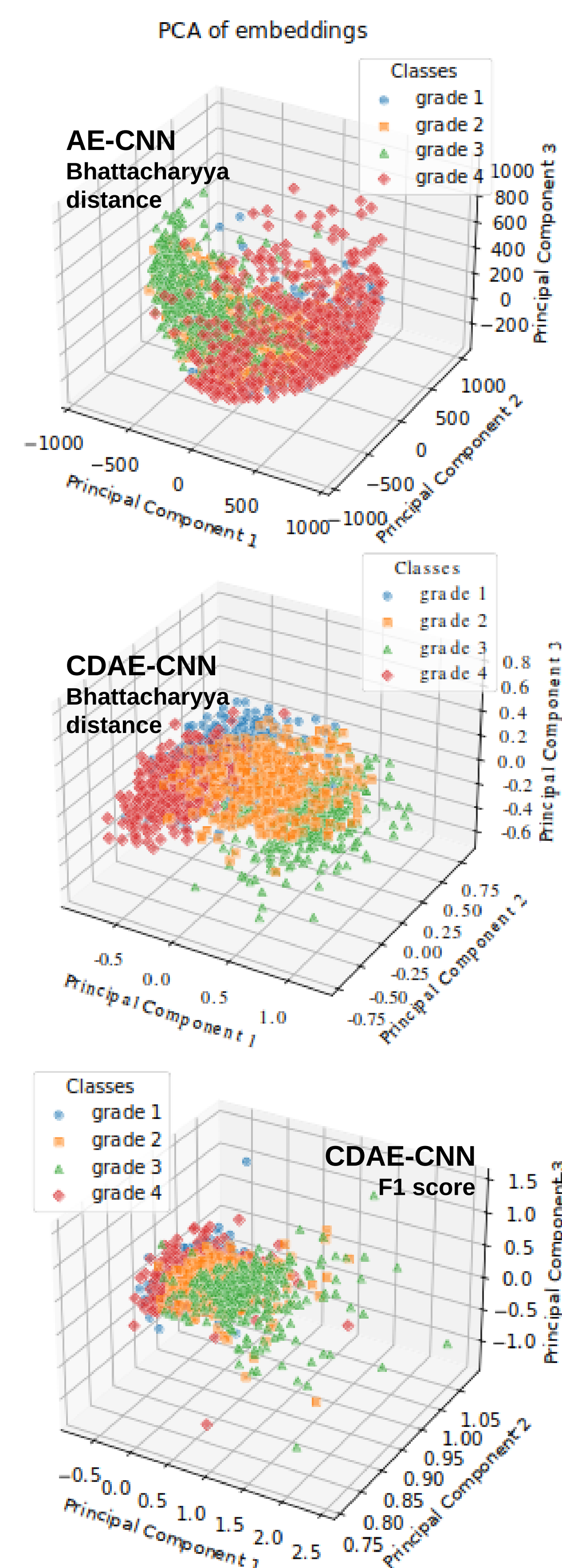
The sketch on the left illustrates the supervised autoencoder scenario, where a classifier branch is added to the latent space. When Bhattacharyya distance is used as the optimization metric, as shown in the topmost sketch, the model adjusts its parameters to maximize class separability in the latent space. In contrast, optimizing based on F1 score guides the model to improve classification performance, which may result in less distinct latent space clustering. The table above compares Optuna optimization results for the CDAE-CNN model using Bhattacharyya distance versus F1 score.

CONCLUSION & DISCUSSION

Our findings confirm that incorporating a **supervised classifier branch (CDAE)** into the autoencoder significantly enhances nuclei classification in clear cell renal cell carcinoma, outperforming **unsupervised variants (AE, CAE, DAE)** in distinguishing critical tumor grades. Although optimization for latent clustering (Bhattacharyya distance) showed strong initial promise, **direct optimization for classification accuracy (F1-score)** yielded better classification performance. This approach offers a practical pathway towards reducing diagnostic uncertainty and manual annotation burden.

Model	AE	CAE	DAE	CDAE
MLP	14.7	24.5	17.43	34.62
CNN	16.33	19.21	25.93	47.23

The table above presents the Bhattacharyya distances of the best-performing models identified by Optuna after architecture search and fine-tuning. The diagrams on the left display pair plots of the first PCA components of the latent space embeddings from the top-performing autoencoders, each optimized for Bhattacharyya distance across four different architectures.



Above is the visualization of the first three PCA components of the training embeddings for the highest-performing models. The top and middle plots show AE and CDAE-CNN respectively, both optimized using Bhattacharyya distance. The bottom plot shows CDAE-CNN optimized using F1 score, as depicted in the left sketch.

